

Day 99

11/07/2025

cloud storage

* cloud storage is service that allow you to store, access and manage data over the Internet, instead of on your local computer or hard drive.

- * 90% of software companies use cloud.
- * Nearly \$600 Billion in End-user spending

Auto Scaling

* Auto Scaling is a cloud computing feature that automatically increases or decreases resources (like servers or containers) based on the current load, traffic or usage.

Before the cloud

- * Buy server
- * install
- * maintain
- * Replace it when it's become old
- * Have own IT team.

on-premise servers

* on premise servers are physical servers that are located within a company's own facilities or data center. These servers are owned and maintained by the organization and are responsible for running all internal applications, database and other IT services.

Key Features :-

- * ownership = organization owns
- * control = Full
- * security = own responsible
- * cost = High
- * maintenance - requires

Cloud servers

* cloud servers are virtualized servers hosted and managed by a "cloud service providers" (eg: AWS, Azure, Google Cloud). These servers are hosted in off site data centers and can be accessed over the internet. Cloud providers are responsible for managing the hardware, networking and physical security.

* Scalability : highly Scalable.
* we can increase or decrease resource based on demand.

* Cost : Typical operates on pay as-you-go - model.

* Flexibility : without worrying about physical infrastructure we can deploy application

* maintenance : handle by cloud providers.

MICROSOFT AZURE

→ 2008 - Announcement.

↳ called "Windows Azure" at
Microsoft Professional Developers
Conference (PDC) in October
2008.

→ 2010 - Public Launch.

↳ become commercially available
↳ offered core services like.

- * Compute

- * Storage

- * SQL Azure (Azure SQL)

- * AppFabric

(middle ware platform for
connectivity)

→ 2014 - Rebranding to Microsoft Azure

↳ Azure began expanding
support for Linux,
Open-source tool

→ 2015 - 2018 - Rapid expansion

→ Data centers across the globe
→ services like Azure machine learning, IoT Hub and Cosmos DB.

→ 2019 - Hybrid cloud & security focus.

→ Launched Azure Arc for hybrid / multi-cloud management.

→ 2020 - 2023 - AI, Analytics & DevOps

→ AI and analytics services
* Azure Cognitive Services
* Azure Synapse Analytics
→ Azure DevOps; GitHub Actions

→ 2023 - 2024 - AI Leadership & Open AI
→ exclusive cloud provider for Open AI

Cloud service models

* core cloud service models

* IaaS - Infrastructure as a service

* PaaS - Platform as a service

* SaaS - Software as a services

IaaS:-

* provides virtualized computing resources like servers, storage and networking over the internet.

eg:- * AWS EC2, Azure VM, google compute engine.

Paas:-

* Offers hardware and software
(eg: runtime, database, Framework)
needed for application development

eg:-

* Google App Engine, Heroku,
Azure App Service.

SaaS

* Delivers software applications
via the Internet typically on a
subscription basis.

eg:-

* Gmail, Microsoft 365,
Drop Box.

Extended Cloud services

* FaaS

↳ Function as a service

* CaaS

↳ Container as a service

* DaaS

↳ Desktop as a service

* BaaS

↳ Backend as a service

* DBaaS

↳ Database as a service

* SaaS

↳ Storage as a service

* MaaS

↳ machine learning as a service

* NaaS

↳ network as ~~example~~ a service.

* AaaS

↳ Artificial Intelligence as a service.

* IDaaS

↳ Identity as a service

* ITaaS

↳ IT as a services.

* XaaS

↳ Everything as a services